Infinite count-controlled Semaev families and the capacity bridge

2026-11-09. Reviewed written existence arguments and fixed software controls.

For $z=a_6=1$, the normalized equation is $x^2y^2+xy+x^2+y^2+1=0$. A direct point correspondence with $E: Y^2+xY=x^3+1$, Hasse's bound and exact subspace probabilities establish the following results.

Exactly two solutions: every odd $n \geq 3$

Some dim-$(n+1)/2$ subspace containing 1; basis selection existential

UNSAT with nonzero restricted trace: infinitely many odd n

Avoids the single constant trace contradiction; other easy proofs may remain

Checked quotient rank q exceeds valid solution capacity U

Direct closure check + count dimension argument implies LFD >= D+1

Growing LFD remains a separate obligation

Order-four target; no prime-order-target or efficient-basis theorem

Why the SAT existence proof works: All full-field zeros except (0,1),(1,0) have 1,x,y linearly independent over F2 for odd n. Averaging over subspaces containing1 gives an expected count below four. Every count is even and at least two, so some subspace has exactly two zeros. This does not certify an arbitrary input subspace.

Why the UNSAT refinement helps: Removing trace-zero subspaces before averaging yields a trace-active UNSAT subspace whenever the elliptic Frobenius trace is nonnegative. The irrational Frobenius angle supplies infinitely many suitable odd extensions. No growing closure obstruction has been constructed.

Implemented bridge: A bounded CLI reruns direct kernel verification, reconstructs structural solution capacities and requires strict q>U. Six software controls pass; the known two-solution control remains inconclusive at q=U. No positive scientific lower-bound fixture was exercised. Code and proof hashes are bound in results. No scientific trial, solver search, native build or cloud job ran.

Evidence: low-solution-count-family.md and its review; trace-active-unsat-family.md and its review; capacity-bridge-audit.md; capacity-bridge-validation/receipt.json. Hasse source: MIT 18.783 (2023), Lecture 7, Theorem 7.3. These are written proofs, not Lean-certified theorems or a novelty claim.